

Assignment Report

Volume Control by hand gestures using Python

Topic: Machine learning

BY

Presha Vaishnav

Table of Content

DATE	CONTENT	PAGE NO.
11/03/2023	Abstract	03
11/03/2023	Introduction	04
11/03/2023	Working Principle	05
11/03/2023	Implementing of project	06
11/03/2023	Methodology	07
11/03/2023	Result	08
11/03/2023	Images	09 to 14
11/03/2023	Conclusion	15
11/03/2023	References	16

Abstract

Computer interaction is one of the main challenges that old age and disabled people face in today's digitalized world. In the last few years, hand gesture recognition has surfaced as one of the most natural human-machine communication in software development, particularly for friendly and flexible human-computer interaction. This report presents volume control using your hand gesture on VS CODE with a camera programmed in Python using the **OPEN CV, MEDIAPIPE AND PYAUTOGUI** libraries.

The hand gesture recognition system's main objective is to make an easy pathway of communication between human and digital systems to control volume via using their hands. The system is useful for operating in real-time for any person using it.

A system that can accept hand movement in real time is hand gesture recognition. The design of hand gesture recognition is quite difficult, as it combines two main problems. The detection of the hand is the first step in creating a sign that can only be manoeuvre by one hand at a time.

It can be used in various places, including human-computer intercommunication, sign language and many more. This report talks about a way for hand gesture detection by using python language. It comprises of a single camera that captures the user's hand gesture and feeds it to the computer.

The primary goal of this gesture recognition project is to develop a system that can recognize specific human hand gestures and make use of them to send information to the device to control the volume of the system. With this method, a user can operate a system by making some specific gesture in front of its camera. With the help of the **Open CV, MEDIAPIPE AND PYAUTOGUI** modules, we will create a volume control by hand gestures. Without the use of a keyboard or mouse, the computer's volume can be controlled via hand gestures.

Keywords- Hand gesture, volume control via hand gesture, opencv, mediapipe documentation.

Introduction

Hand gesture detection helps computers to understand human body sign language. This helps to build a pathway between humans and computers, rather than just the basic texts. In this report for hand gesture detection, the human body's movements are read by a systems camera. The main objective of this project is to develop a communication which will capture hand gestures dynamically and control the volume level through it.

Open CV: - Open CV is a library of python. It is used in various projects to detect the hand and face and is also used in other programming languages. OpenCV is a large open-source library for the computer, image processing etc. By using this, one can process images, videos and even the handwriting of a human being.

Pyautogui:- On Windows, Pyautogui accesses the window to give output through the built-in modules.

Mediapipe:- It is an open-source library, which has some solutions for hand gesture detection and recognition and provides a summarization of python. MediaPipe Hands is a high-precision hand-tracking solution. It uses ML to infer 21 key 3D hand information from just a single frame. We can use it to extract the co-ordinates of the given key points of any hand.

With the help of all these modules our program will work or you can use other libraries and modules if you want.

Now importing all the modules in the VS code to design the algorithm for the project.

Working Principle

The web camera of the computer is used in this design. It detects and recognizes hand gestures if shown in front of the frame/camera. Some landmarks are provided for the detection of hand gestures.

When the user brings one hand in front of the frame/camera the landmarks specified for each finger will start showing on the screen.

In this project, we have used the points of the tip of the index finger and the tip of the thumb. The points given to both according to mediapipe documentation are 4 and 8 respectively.

When a hand is detected and recognized by the camera the landmarks will be seen on the palm. As here we have taken the index finger's tip and thumb tip, we will highlight their points to make them stand out and help us differentiate which points are appointed to them.

As of here, this design works like:

Step 1: -Open VS code or whichever software you prefer.

Step 2: - Import all the modules libraries OpenCV, mediapipe and pysutogui.

Step 3: - Write the algorithm using these libraries.

Step 4: - Executing the while loop to detect the hand landmark.

Step 5: - The main aim of this is to detect the hand gestures and recognize them.

Step 6: - A display window will show on the screen. Bring your hand in front of it and some dot like landmark will start showing on your palm.

Step 7: - Using these hand gestures one can control the volume of the system.

Implementing the project

This report is written and executed in python language using VS code interpreter.

Importing the libraries from mediapipe, open cv and pyautogui in VS code for the working of the project. Open cv is used for reading the image shown in front of the frame/camera. Mediapipe is used for processing intuitive data for audio, video, hand gestures and many more. There are various solutions given in mediapipe for different types recognition. Here we will use the Mediapipe hand for hand gesture recognition. Pyautogui help find the image given/shown to the camera and on the screen. It gets the distance and coordinates or shape for the computer.

By importing all these modules, it will help us design our algorithm properly without any trouble. Using mediapipe library we will get proper hand gesture solution points shown on the palm of our hand for controlling the volume.

By applying all these together and making an algorithm we will see the result and how to works.

Using the conditional statement in the algorithm for the recognition of the hand gestures if present any at the given time. And, to see if there is single hand or multiple hands present in the frame/camera. Transfiguring the image to RGB image and flipping it so the user can see it properly. Declaring the points given to the index finger and the thumb that has to be used (8 and 4 respectively). Managing the distance between both of the tips for controlling the volume through them. Highlighting the points of the tips which we will use to control the volume by drawing different colour circles around them. Using algorithm to draw the line between the two points given which are -

Tip of index finger: 8

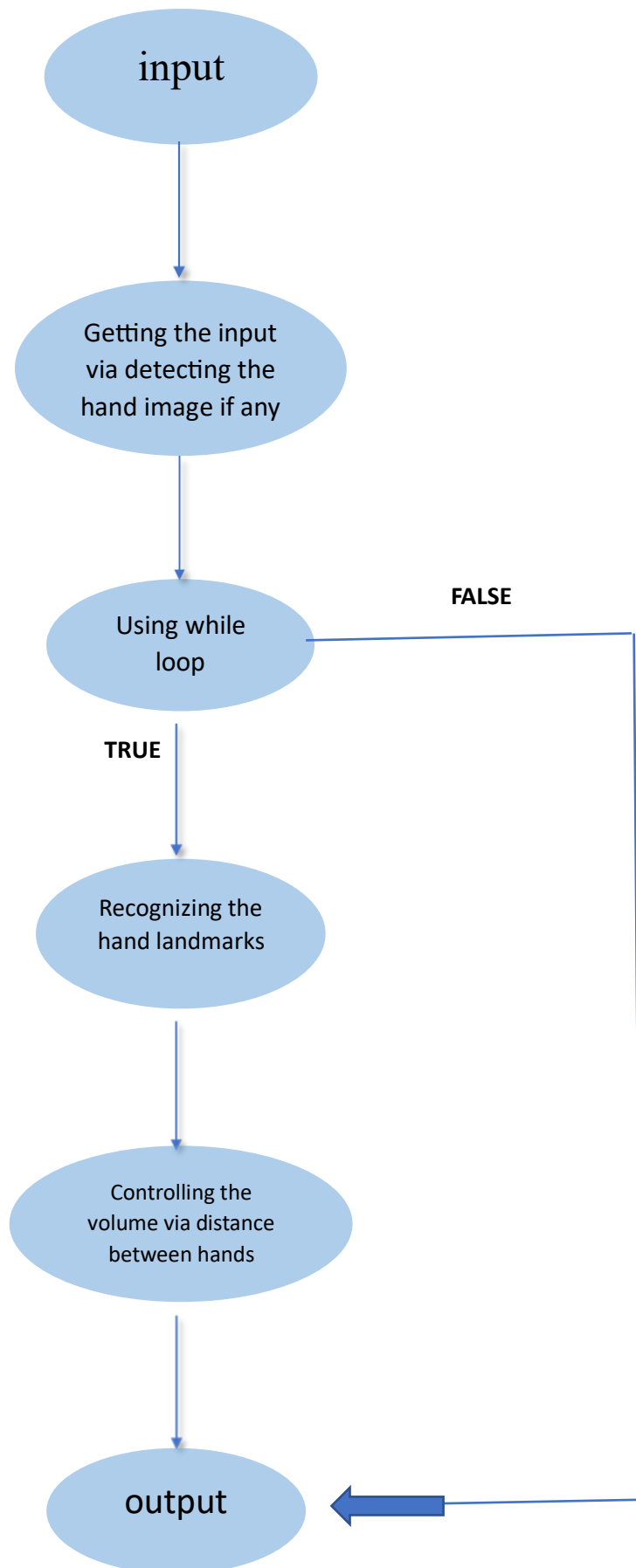
Tip of the thumb: 4

Now after calculating the distance, we will convert the hand distance to the volume range and manage the hand gestures for maximum volume and minimum volume.

And then image/video window will display the output.

In the end, we will terminate the program.

Methodology



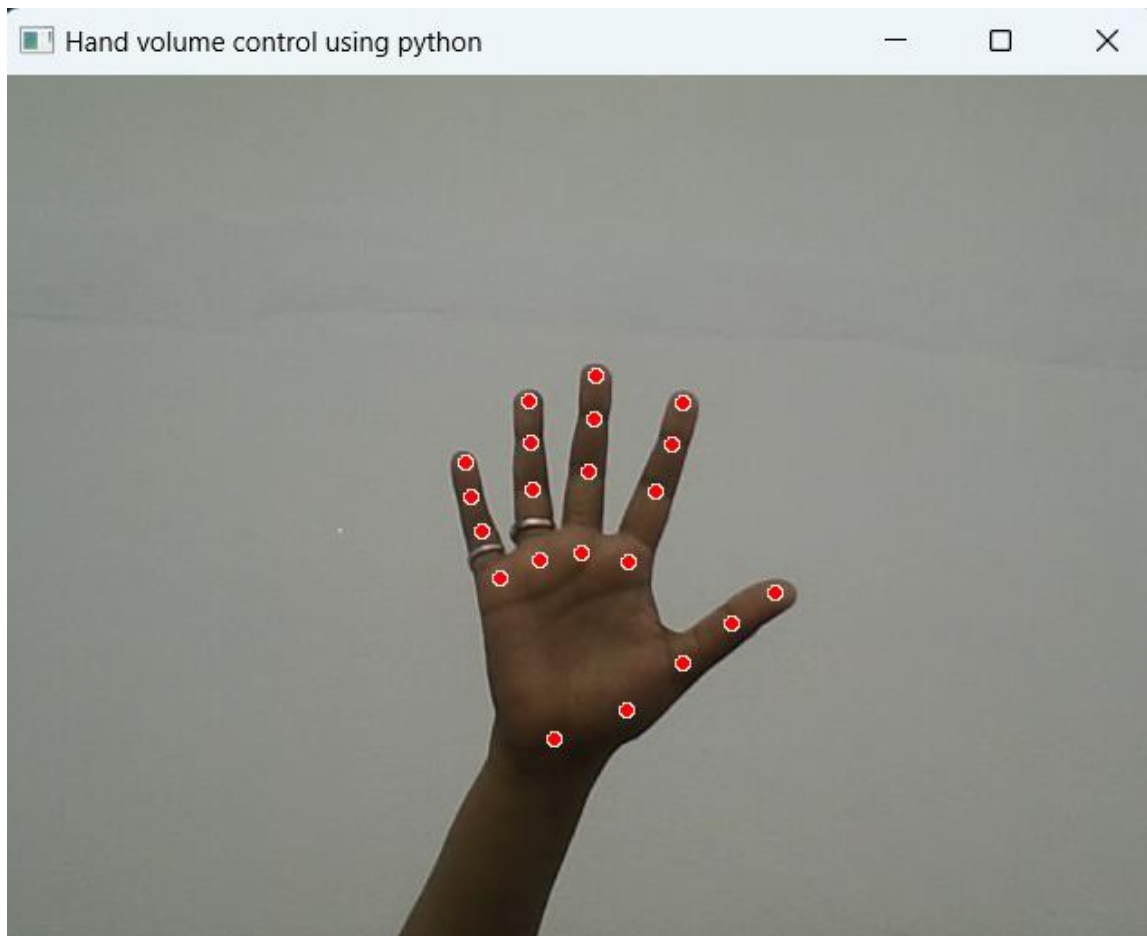
Result

The main objective of this report is volume control via hand gestures using the camera of the computer.

When we run the algorithm, we can see the change in the volume depending upon the distance between the tip of the index figure and the tip of the thumb.

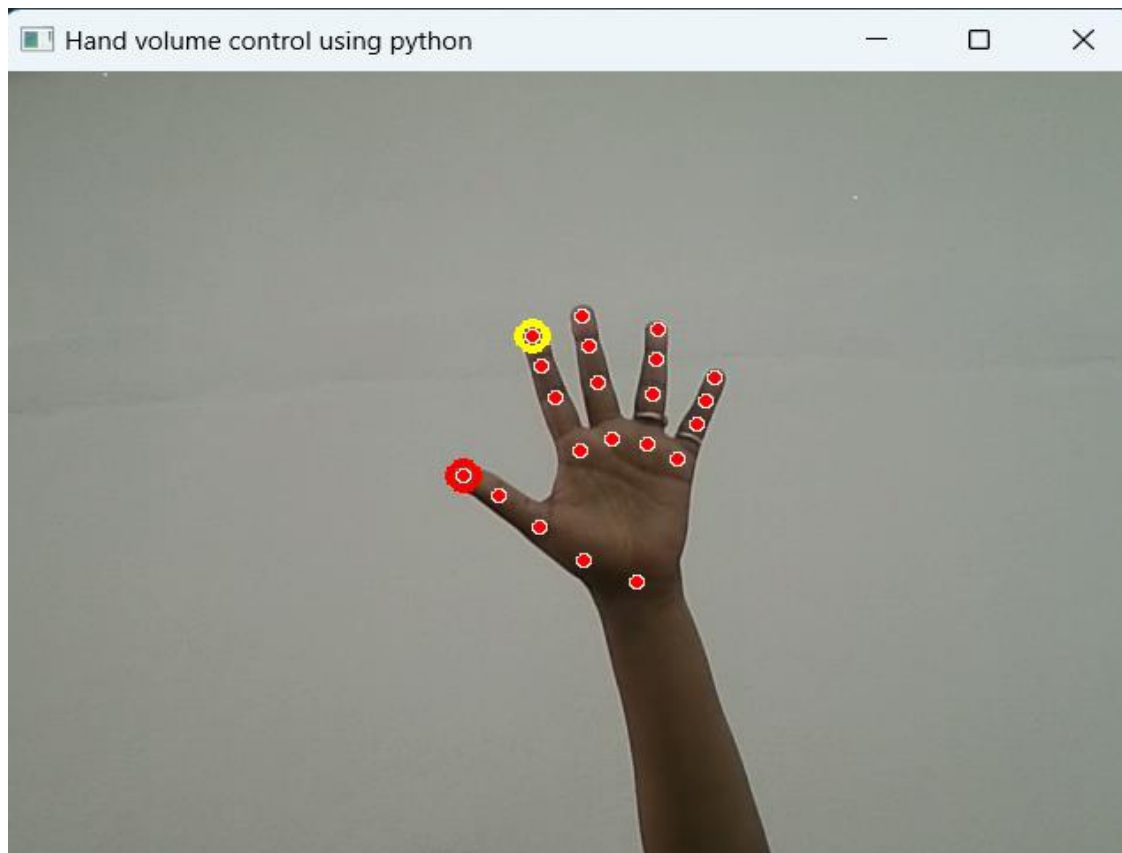
The below images are the output of the algorithm which runs without any error.

IMAGE 1



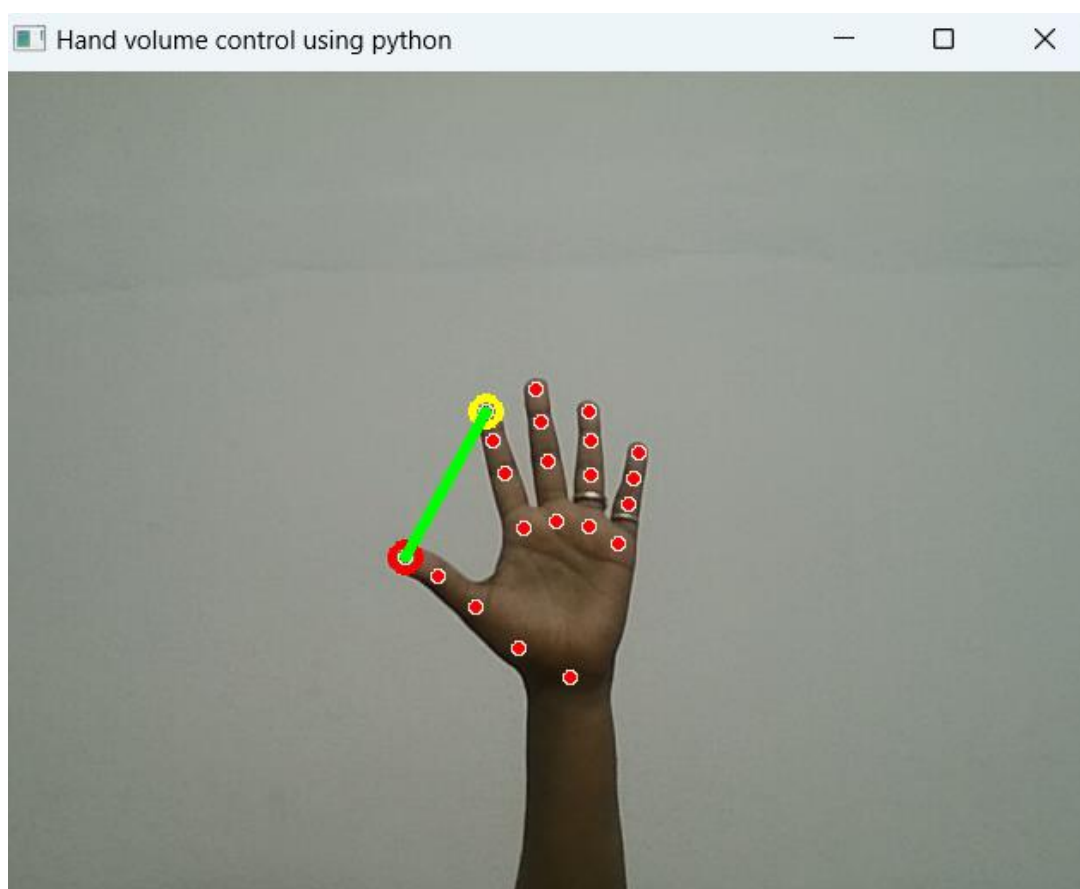
Dots-like points can be seen on the palm which shows that the hand-detection part of the algorithm works perfectly fine.

IMAGE 2



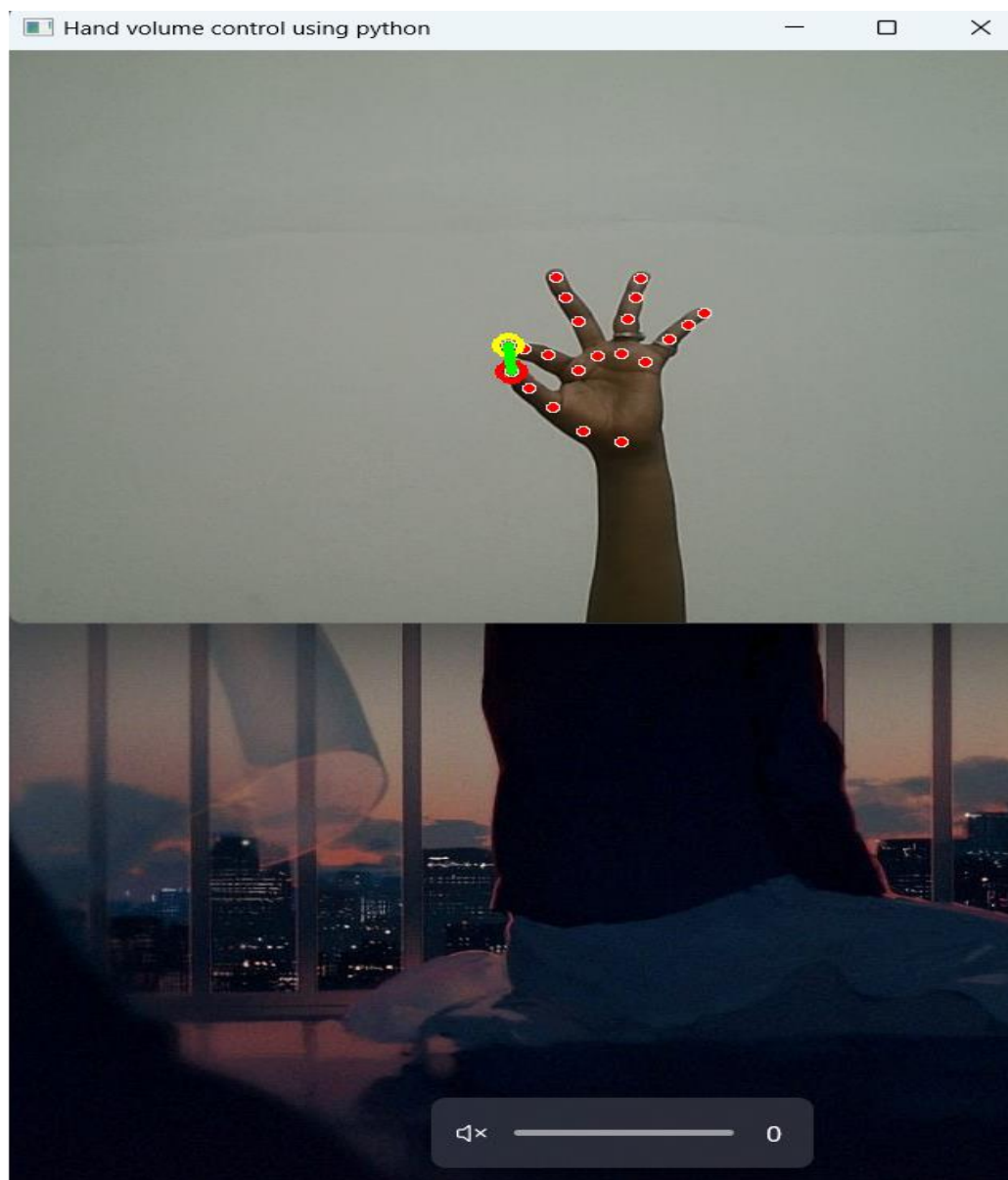
The above image shows the highlighted area of the index finger's tip and thumb tip. These parts are used for controlling and managing the volume of the system.

IMAGE 3



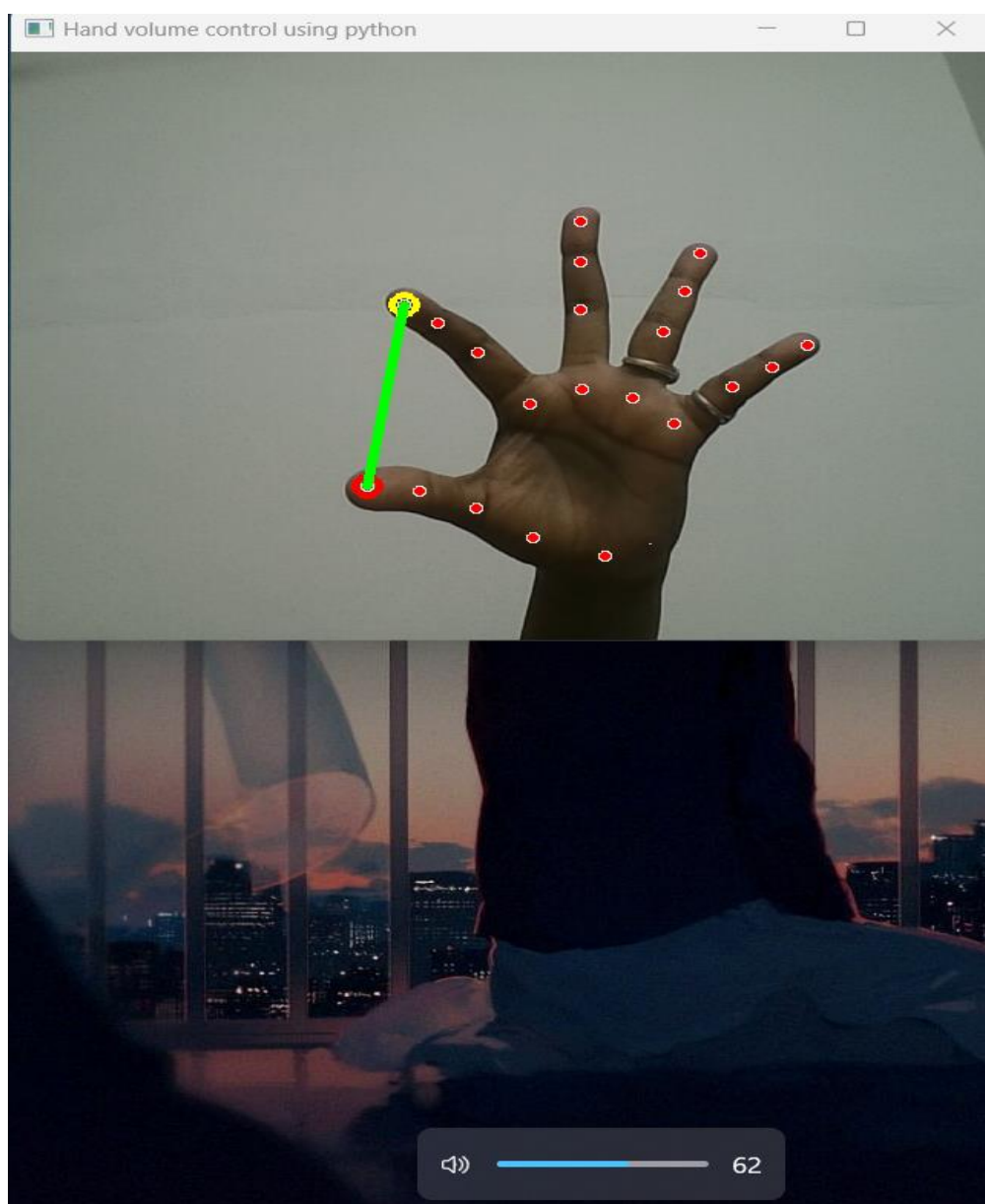
This image shows a green colour line which cover-ups the distance between the tip of the index figure and the tip of the thumb. The volume controlling is depended on the length of this line.

IMAGE 4



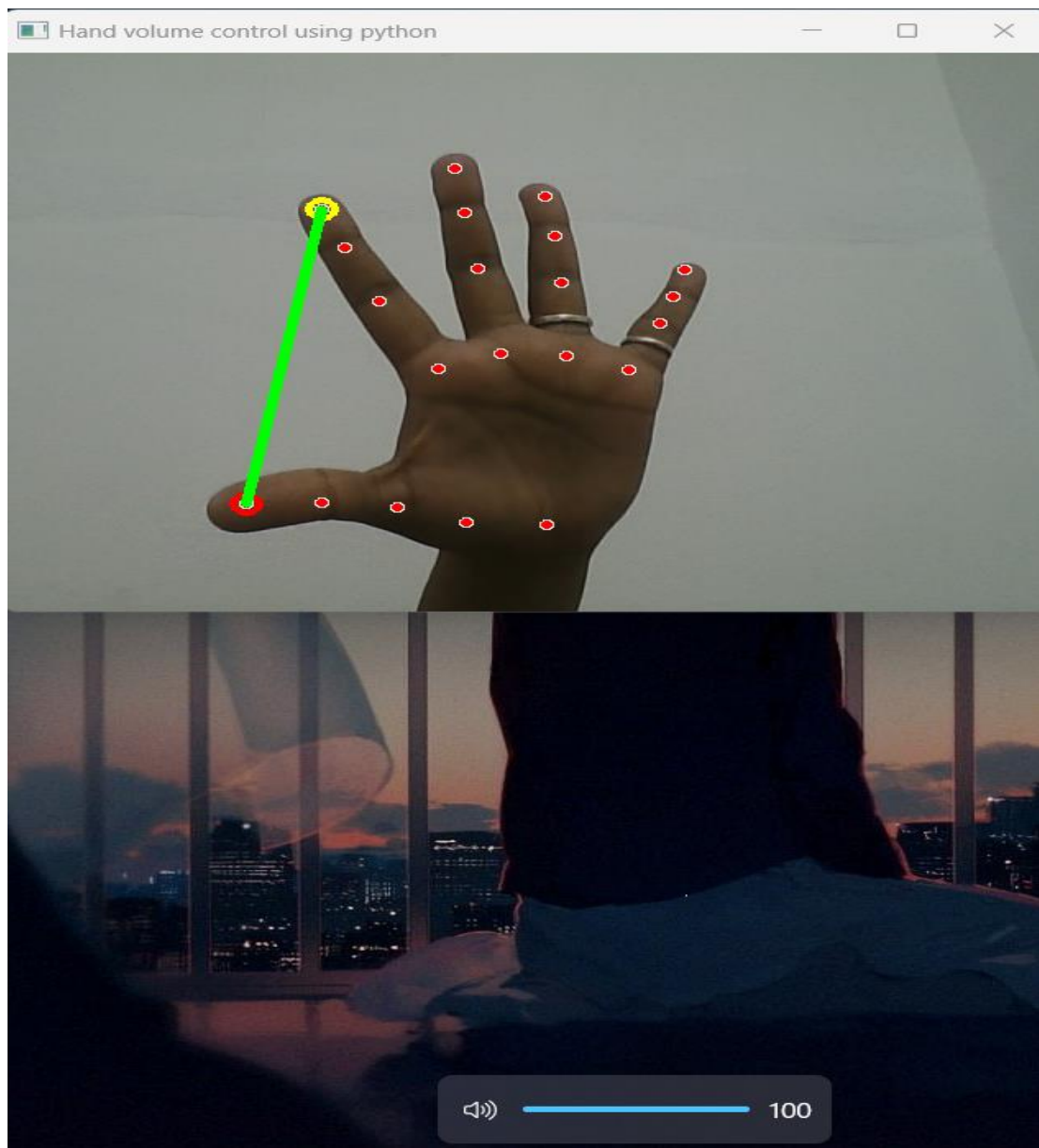
As shown in the above image the distance between the tip of the index finger and the thumb is the minimum and so the volume of the system is zero as seen.

IMAGE 5



As shown in the above image the gap/distance between the tip of the index finger and the tip of the thumb is almost half, therefore the volume is around 50 to 70.

IMAGE 6



As seen in the above image the distance between the tip of the index finger and the tip of the thumb is maximum, hence the volume's at the maximum which is 100.

Conclusion

The main objective of this project is to make a system that controls the volume of the system via hand gestures. This report talks about how it works and the algorithm used behind it. This hand gesture-based project helps one to handle the volume of the computer in a much more suitable way. In this project particularly the use of the tip of the index finger and the tip of the thumb detect and recognize and then controls the volume of the device.

In this project, as seen above the algorithm runs perfectly fine and the volume is controlled through it in a much easier way.

The primary focus of this project is to digitalize and make it trouble-free for users to use any electronic devices. This project is a plausible and effortless way to control the volume of systems and devices without much physical work. Users do not have to wear any special commodity to operate it, it can be operated with bare hands.

Also, it can work on a low-budget camera. One does not have to pay any extra penny.

With the help of open cv, pyautogui, and mediapipe libraries we have gained 90% of precision in this project of volume control using hand gestures. The important part used and needed for this project is the web camera.

This whole project is executed on VS code using python language. A very favourable way of handling the volume of the system.

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